

[Pranesh S A 251801234-2025-AIDS](#) ▾**P2****Started on** Thursday, 20 August 2026, 8:39 AM**State** Finished**Completed on** Thursday, 20 August 2026, 8:42 AM**Time taken** 2 mins 43 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, count = 0;
5      scanf("%d", &n);
6
7      int i = 1;
8      count++;
9      int s = 1;
10     count++;
11
12     while(s <= n) {
13         count++;
14         i++;
15         count++;
16         s += i;
17         count++;
18     }
19     count++;
20
21     printf("%d", count);
22     return 0;
23 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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